Chain Stitch Machine



Category: **Sewing, Textile & Handicraft** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Chain stitch machine

What it is used for

A chain stitch machine forms interlocking chain-shaped stitches to join or decorate fabric in garment and textile production. Correct threading, stitch settings, and fabric feeding help produce consistent seams while reducing needle injuries and thread breakage.

Main parts

- Machine head and sewing needle
- Looper or chain-forming mechanism
- Thread stand and thread guides
- Tension controls and stitch-length regulator
- Presser foot and feed dogs
- Motor, foot pedal, and protective guards

Before you start

1. Confirm the fabric type, seam requirement, thread type, and required stitch length.
2. Inspect the needle, presser foot, looper, thread guides, guards, and power cable for damage.
3. Fit the correct needle and confirm that it is securely mounted.
4. Thread the machine according to its threading diagram, including the looper where applicable.
5. Adjust thread tension, stitch length, and differential feed if fitted, according to the fabric specification.
6. Test the stitch on a fabric scrap and check seam formation, tension, and thread security.

How to use

1. Switch off the machine before changing the needle, threading near moving parts, or clearing a jam.
2. Raise the presser foot and position the fabric beneath it at the starting point.
3. Lower the presser foot and ensure the fabric is aligned with the intended seam path.
4. Start the machine using the approved control and guide the fabric gently without pulling or pushing it forcibly.
5. Maintain a steady sewing speed and keep the fabric flat as it passes through the feed mechanism.
6. Observe the stitches for consistent loops, correct tension, and even spacing.
7. Stop the machine if the thread breaks, stitches skip, fabric jams, or unusual noise occurs.
8. Wait for the needle and moving parts to stop before adjusting the fabric or removing tangled thread.

9. At the end of the seam, follow the machine's approved method for securing or finishing the chain stitches.
10. Remove the fabric carefully and inspect the seam for skipped stitches, loose loops, and uneven tension.
11. Switch off the machine and clean lint and thread waste from the needle plate and feed area.
12. Report persistent stitch defects or mechanical problems before further production.

⚠ Safety

- Only trained operators should use or adjust the machine.
- Keep fingers away from the needle, looper, and feed mechanism while the machine is running.
- Do not bypass needle guards or other safety devices.
- Switch off and isolate power before replacing the needle, clearing jams, or performing maintenance.
- Keep loose clothing, hair, and jewellery away from moving parts.
- Use the correct needle and never operate the machine with a bent or damaged needle.

Care & storage

- Remove lint and thread waste after shutdown, especially around the feed dogs and looper area.
- Inspect the needle, looper, presser foot, and thread guides regularly.
- Lubricate only designated points according to the manufacturer's instructions.
- Check thread tension and stitch formation at the start of each production run.
- Arrange servicing if the machine repeatedly skips stitches, produces abnormal noise, or breaks thread.

Common mistakes to avoid

- Threading the looper or needle incorrectly.
- Using incorrect thread tension or stitch length for the fabric.
- Pulling the fabric forcefully while sewing.
- Adjusting the needle or clearing tangled thread while moving parts are active.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

Prepared by - Subodh Kumar